

LECTURE 2

Supervised Learning Part 1: Classification Algorithms for Defence

Logistic Regression • kNN • Decision Trees • Random Forest • SVM • Naive Bayes

Case study: 5-class IADS threat classification, N = 8,500 samples, 11 radar + EW features

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What we cover today: 6 classifiers, 1 unified dataset

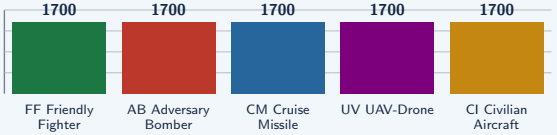
Learning objectives

1. Derive decision boundaries for key classifiers
2. Implement logistic regression mathematically
3. Understand bias–variance for kNN (k effects)
4. Build & prune a decision tree from first principles
5. Explain random-forest bagging & feature importance
6. Derive SVM hard & soft margins; the kernel trick
7. Apply Naive Bayes to a classification problem
8. Select the right classifier for a defence scenario
9. Compute precision, recall, F1, ROC-AUC from scratch
10. Identify class imbalance & apply fixes

Unified case study protocol

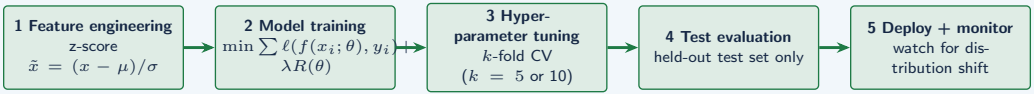
Each algorithm is trained on the SAME data and compared on accuracy, F1-macro, per-class precision/recall, training time & inference latency, and decision-boundary geometry.

5-class IADS threat dataset: $N = 8{,}500$, 11 features, split 68/17/15



Feature	Feature	Feature
x_1 Peak frequency (GHz)	x_5 Scan rate (deg/s)	x_9 EW jamming (boolean)
x_2 Pulse width (μs)	x_6 Polarisation (categ.)	x_{10} Course (deg)
x_3 PRF (Hz)	x_7 Doppler shift (MHz)	x_{11} Speed (km/h)
x_4 Amplitude (dBm)	x_8 Sidelobe level (dBc)	

Supervised classification: the general pipeline



Setup

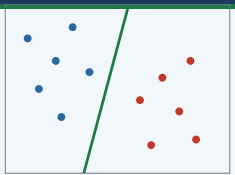
Training set $D = \{(x_i, y_i)\}_{i=1}^{N}$, $x_i \in \mathbb{R}^d$ with $d = 11$, $y_i \in \{$FF, AB, CM, UV, CI$\}$. **Never** peek at the test set during training.

DATA-PITFALL CHECK before training: ① class balance ② feature scaling applied ③ no data leakage (future information in training) ④ categorical features encoded ⑤ missing values handled.

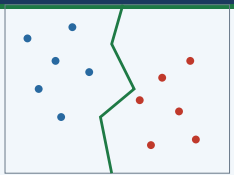
Classifier	Type	Parametric	Needs scaling
Logistic reg.	Linear	Yes	No
kNN	Instance	No	**YES**
Decision tree	Non-linear	No	No
Random forest	Ensemble	No	No
SVM	Margin	Yes	**YES**
Naive Bayes	Probabilistic	Yes	No

Red = mandatory feature scaling; green = scaling not required.

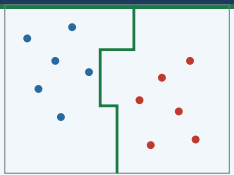
Decision Boundaries — Each Classifier Carves Space Differently



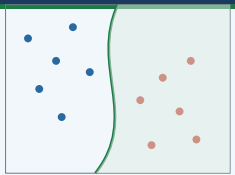
Logistic regression: hyperplane $w \cdot x + b = 0$



kNN: Voronoi, local



Decision tree: axis-aligned staircase

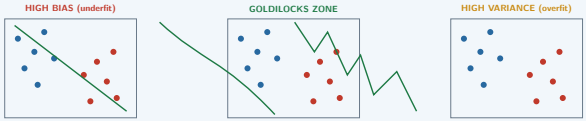


SVM / RBF: smooth max-margin

- **Logistic:** works when classes are linearly separable
- **kNN:** any shape; noisy near boundaries
- **Tree:** interpretable but straight-edged only
- **SVM/RBF:** optimal generalisation via kernels

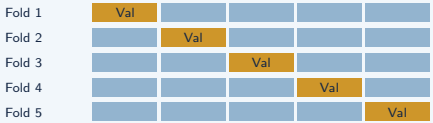
IADS context: in our 11-D feature space a "decision boundary" is a hyperplane or hypersurface separating 5 class regions. Linear = 5 hyperplanes (one-vs-rest); RBF-SVM finds a curved separating surface.

Bias–variance trade-off



- **High bias:** too simple, wrong even on training data (straight line for a curved boundary; LR on non-linear data)
- **High variance:** perfect on train, fails on test (deep tree, 1-NN)
- **Goldilocks:** low bias + low variance — found by cross-validation

Evaluation protocol: train → tune → test



Never use test data for tuning! Average over the 5 folds: a robust estimate of generalisation performance before the one-time final test evaluation.

MODEL (BINARY; EXTEND BY OVR / SOFTMAX)

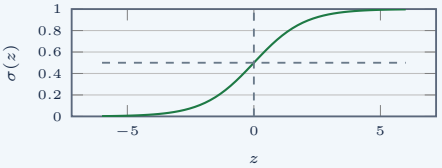
$$z = w^\top x + b, \qquad \sigma(z) = \frac{1}{1+e^{-z}} \in (0,1)$$

$$P(y{=}1|x) = \sigma(w^\top x + b)$$

$$\sigma(z) \geq 0.5 \iff w^\top x + b \geq 0 \quad \text{(a hyperplane)}$$

MULTI-CLASS SOFTMAX (5 WEIGHT VECTORS)

$$P(y{=}c|x) = \frac{\exp(w_c^\top x)}{\sum_k \exp(w_k^\top x)}$$



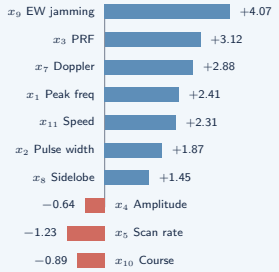
IADS results (OvR, $N = 8{,}500$)

Class	Precision	Recall	F1
FF (Friendly)	0.881	0.903	0.892
AB (Adv. Bomber)	0.896	0.874	0.885
CM (Missile)	0.921	0.908	0.914
UV (UAV)	0.887	0.861	0.874
CI (Civilian)	0.912	0.931	0.921
Macro			**0.897**

- Overall accuracy **89.7%**; training 0.8 s ($N = 5{,}780$); inference 0.09 ms/sample

AB recall = 87.4%: 12.6% of bombers are missed! Is that acceptable for an IADS?

Weights w_j for the AB class (OvR)

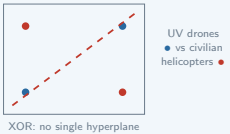


EW jamming (x_9) is the strongest predictor: $w = 4.07 \Rightarrow$ odds ratio $e^{4.07} = 58.5\times$.

ODDS RATIO OF FEATURE j

$$\text{OR}_j = e^{w_j}$$

Where LR fails — and what to use instead

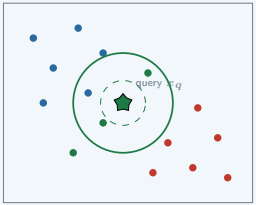


XOR: no single hyperplane

- **Problem 1: XOR boundary.** UV drones fly low/slow, the SAME as civilian helicopters at low altitude $\Rightarrow$ need RBF-SVM or RF
- **Problem 2: interaction.** AB danger = high speed AND high PRF; LR is additive ($w_1x_1 + w_2x_2$) $\Rightarrow$ add x_1x_2 or use trees/SVM
- **Problem 3: non-monotonic.** Supersonic business jets (civilian) have very high speed $\Rightarrow$ RF handles this naturally

Accuracy on non-linear classes: **LR 89.7%** vs **RF 96.8%** vs **SVM 95.4%**.

k-Nearest Neighbours — "You Are Judged by the Company You Keep"



dashed: $k=1$ solid: $k=5$ → majority vote

Algorithm

(1) Euclidean distance to ALL training points: $d(x_q, x_i) = \sqrt{\sum_j (x_{qj} - x_{ij})^2}$ — **normalise first**, distance is scale-sensitive. (2) $N_k(x_q)$ = the k nearest. (3) Vote. (4) Optional distance weighting $w_i = 1/d(x_q, x_i)$.

MAJORITY VOTE

$$\hat{y} = \arg\max_c \sum_{x_i \in N_k(x_q)} \mathbb{1}[y_i = c]$$

Special cases

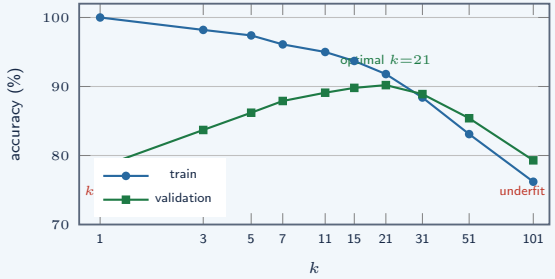
$k = 1$: class of the single nearest neighbour — training accuracy 100%, but overfits. $k = N$: majority class of the whole training set — high bias, ignores local structure. $k = \sqrt{N}$: common starting heuristic ($\sqrt{5780} \approx 76$ here).

Defence considerations

✓ No training phase: updates instantly when new signatures are seen. ✓ Natively multi-class. ✓ Good for rare threat types with few examples.

Inference cost $O(N \cdot d)$: $5{,}780 \times 11 = 63{,}580$ ops/query; 12.2 ms is borderline for IADS ⇒ k-d / ball trees ($O(\log N)$ avg). **Curse of dimensionality**: for $d \gtrsim 20$ all points become equidistant.

kNN — Choosing k by Cross-Validation on the IADS Data



Optimal $k = 21$: validation 90.2%; test accuracy at $k = 21$ is **89.8%**.

k	Train	Val	Inference
1	100%	78.3%	11.2 ms
5	97.4%	86.2%	11.6 ms
11	95.0%	89.1%	12.1 ms
21	91.8%	90.2%	12.5 ms
31	88.4%	88.9%	13.1 ms
101	76.2%	79.3%	16.2 ms

The classic bias–variance U-shape: small k memorises noise; large k over-smooths and ignores local structure. Inference time also grows with k.

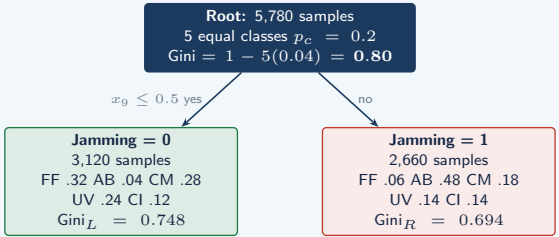
IMPURITY AND INFORMATION GAIN

$$G(\text{node}) = 1 - \sum_c p_c^2 \qquad H(\text{node}) = -\sum_c p_c \log_2 p_c$$

$$IG = H(\text{parent}) - \left[w_L H(\text{left}) + w_R H(\text{right})\right]$$

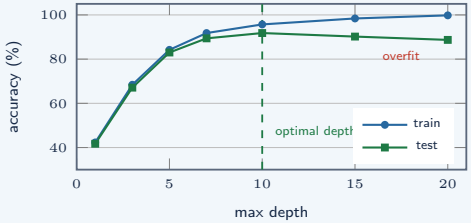
- Greedy, top-down, recursive: at each node search all $d = 11$ features $\times$ all $N - 1$ thresholds; pick (j^*, t^*) with maximum impurity reduction
- **Stop:** max depth; $<$ min_samples (e.g. 5); pure node; no improving feature
- **Post-pruning:** cost-complexity $\alpha \cdot$ number of leaves; choose α by CV

Worked root split (IADS)



Weighted Gini $= \frac{3120}{5780}(0.748) + \frac{2660}{5780}(0.694) = 0.4036 + 0.3194 = 0.723$. Reduction $0.800 - 0.723 = \mathbf{0.077}$: the best split — **EW jamming** is the most informative root feature.

Depth analysis (IADS 5-class)

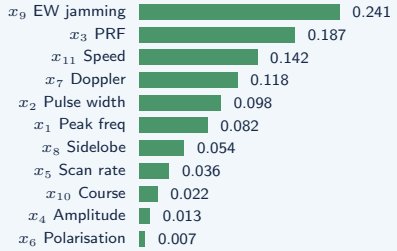


Depth 10: 317 nodes, test 91.8%. Depth 20: 2,103 nodes; unlimited: 5,231 nodes, test 87.4%. **Post-pruning** ($\alpha = 0.002$): 317 $\rightarrow$ 89 nodes (72% fewer), test $91.8\% \rightarrow \mathbf{92.4}\%$ — smaller and better.

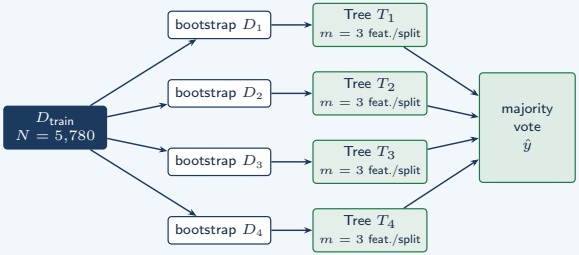
Top decision rules (depth-3 extract)

```
IF Jam=1 AND PRF>2400 AND Spd>650 ⇒ AB (91.4%)
IF Jam=0 AND PW<0.8 AND Scan<2 ⇒ CM (88.2%)
IF Jam=0 AND Spd<250 ⇒ Civilian (93.7%)
```

Feature importance (Gini reduction)



Defence advantage — fully auditable AI: a commander can ask "WHY did the system classify this as an Adversary Bomber?" and receive a sequence of IF–THEN conditions: essential for Rules-of-Engagement compliance. EW jamming is again the single most discriminative feature, consistent with the expert-system finding.



Bagging algorithm

1. Draw B bootstrap samples (with replacement; ≈63.2% unique per sample; the rest are out-of-bag, OOB). **2.** Train B deep trees; at each node consider ONLY $m = \sqrt{d} = \sqrt{11} \approx 3$ random features — this *decorrelates* the trees. **3.** Predict by majority vote.

VOTE AND VARIANCE REDUCTION

$$\hat{y} = \arg\max_{c} \sum_{b=1}^{B} \mathbb{1}[\hat{y}_b(x) = c]$$

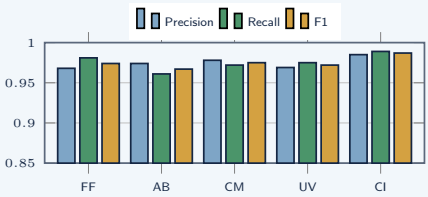
$$\text{Var}_{\text{ens}} = \rho\sigma^2 + \frac{(1-\rho)\sigma^2}{B}$$

- Bias stays the same; variance falls toward $\rho\sigma^2$ — so keep tree correlation ρ low
- **OOB error** ≈ validation error (free!)
- Hyper-parameters: B (100–2000), m ($\sqrt{d}$ or $d/3$), max depth (usually unlimited), min leaf (1–10)

Tuned: $B = 500$, $m = 3$, depth ∞, min leaf 2

Class	Precision	Recall	F1
FF	0.968	0.981	0.974
AB	0.974	0.961	0.967
CM	0.978	0.972	0.975
UV	0.969	0.975	0.972
CI	0.985	0.989	0.987
Macro F1			**0.975**

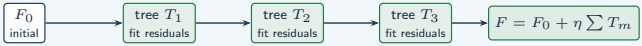
- Overall **97.5%**; OOB error 2.7% (matches test error!)
- Training 12.4 s (500 trees in parallel); inference 2.8 ms



Permutation importance (consistent with a single tree)

EW jamming **0.228**, PRF **0.194**, speed **0.147**.

RF is the accuracy winner for tabular defence classification: robust, low overfit, importances for free.



each tree corrects the errors of the ensemble so far

Boosting vs bagging

Bagging (RF): trees in PARALLEL, independent; reduces VARIANCE; each tree sees ∼63% of data. **Boosting:** trees SEQUENTIAL; each corrects errors of the previous ensemble; reduces BIAS (also variance).

GRADIENT BOOSTING (FRIEDMAN 2001)

$$r_{im} = y_i - F_{m-1}(x_i), \qquad F_m(x) = F_{m-1}(x) + \eta\, T_m(x)$$

$$F(x) = F_0 + \eta \sum_{m=1}^{M} T_m(x)$$

XGBoost improvements

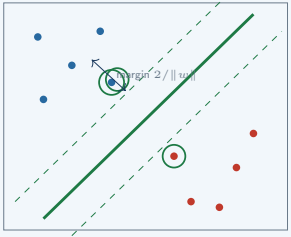
+ second-order Taylor expansion of the loss; + L1/L2 regularisation on tree structure; + column subsampling per split; + parallel (histogram) split finding.

Tuned XGBoost and test results

$M = 400$, $\eta = 0.05$, depth 5, subsample 0.8, colsample 0.8, $\alpha = 0.1$, $\lambda = 1.0$. **Accuracy 96.4%**, macro-F1 0.963, **AB recall 96.8%** (vs LR's 87.4%); training 18.6 s; inference **0.7 ms**.

Vs RF: RF is slightly more accurate (97.5% vs 96.4%); XGBoost has faster inference (0.7 vs 2.8 ms) and a smaller model (better for embedded), but more hyper-parameters. **Rule of thumb:** tabular → XGBoost/RF; raw image/signal → CNN; sequence → LSTM/Transformer.

SVM — Maximum-Margin Classifier: Hard-Margin Formulation



circles = support vectors solid: $w^\top x + b = 0$

HARD-MARGIN PRIMAL

$$\min_{w,b} \tfrac{1}{2}\|w\|^2 \quad \text{s.t.} \quad y_i(w^\top x_i + b) \geq 1 \;\; \forall i$$

DUAL PROBLEM AND SOLUTION

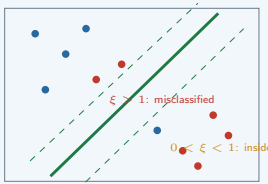
$$\max_{\alpha} \sum_i \alpha_i - \tfrac{1}{2} \sum_{i,j} \alpha_i \alpha_j y_i y_j x_i^\top x_j$$

$$w^* = \sum_i \alpha_i^* y_i x_i, \qquad \hat{y} = \text{sign}\Big(\sum_i \alpha_i y_i x_i^\top x + b \Big)$$

Why maximum margin matters in defence

Many hyperplanes separate AB from FF signals; SVM picks the ONE farthest from both clusters — the most robust to sensor noise, slight adversary variation in signature, and PRF/frequency measurement error.

- ✓ only support vectors matter (sparse); unique optimum (convex QP)
- ✓ depends only on dot products $x_i^\top x_j$ ⇒ **kernel trick**
- × fails if not linearly separable ⇒ soft margin

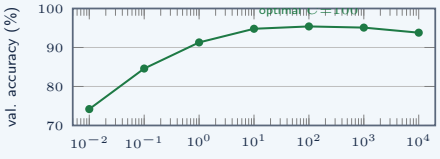


SOFT-MARGIN PRIMAL (SLACK $\xi_i \geq 0$)

$$\min \ \tfrac{1}{2}\|w\|^2 + C\sum_i \xi_i \quad \text{s.t. } y_i(w^\top x_i + b) \geq 1 - \xi_i$$

- $\xi_i = 0$: correct, outside margin; $0 < \xi_i < 1$: inside margin, correct side; $\xi_i > 1$: misclassified
- Small C: more violations, wide margin, may underfit; large C: narrow margin, may overfit; $C = \infty$: hard margin

Tuning C (IADS, RBF kernel)



Class	Prec	Rec	F1
FF	0.961	0.973	0.967
AB	0.967	0.952	0.959
CM	0.971	0.965	0.968
UV	0.958	0.961	0.960
CI	0.978	0.984	0.981

Macro F1 0.967; overall 96.7%; 1,247 SVs (21.6% of training data); training 38.4 s; inference 4.1 ms

Defence implication: with asymmetric costs (missing an AB is worse than a false alarm) use class-weighted C: $C_{AB} = 5 \times C_{\text{others}}$. Penalty is not equal across all violations.



KERNEL FUNCTION

$$K(x_i, x_j) = \phi(x_i)^\top \phi(x_j)$$

- Dot product in a higher-dimensional space **without** ever computing $\phi(x)$
- **Linear:** $x^\top x'$ **Polynomial:** $(x^\top x' + c)^d$; $d = 2$ maps $\mathbb{R}^{11} \to \mathbb{R}^{78}$
- **RBF:** $\exp(-\gamma \|x - x'\|^2)$; γ small = smooth, large = sharp

RBF grid search: validation accuracy (%)

	C=1	C=10	C=100	C=1000
γ=0.001	81.2	83.7	85.2	85.3
γ=0.01	88.4	91.2	93.8	93.9
γ=0.08	91.3	94.8	95.4	95.1
γ=0.5	88.7	90.1	92.3	91.8
γ=2.0	74.2	75.8	76.4	76.1

Optimum $\gamma = 0.08$, $C = 100$: **95.4%**. Large $\gamma = 1.0$: 74.1% (too local, overfit); small $\gamma = 0.01$: 89.2% (too smooth).

Mercer & defence-specific kernels

Valid iff K symmetric PSD. **Spectral:** $\exp(-\|\text{FFT}(x) - \text{FFT}(y)\|^2 / 2\sigma^2)$. **DTW:** $\exp(-\text{DTW}(x, y)/\sigma)$ — speed-invariant tracks. **String:** shared k-grams for SIGINT intercepts.

Naïve Bayes — Prior × Likelihood = Posterior

BAYES' THEOREM + INDEPENDENCE ASSUMPTION

$$P(y{=}c|x) \;\propto\; P(y{=}c)\prod_{j=1}^{d} P(x_j|y{=}c)$$

$$P(x_j|y{=}c) = \mathcal{N}(x_j;\mu_{cj},\sigma^2_{cj})$$

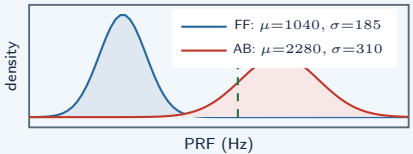
$$\hat{y} = \arg\max_c \Big[\log P(y{=}c) + \sum_j \log P(x_j|y{=}c)\Big]$$

- "Each feature is conditionally independent given the class" — almost never true, yet often works
- Training = class means & variances: $O(Nd)$; ≈0.03 s for IADS

Test results (Gaussian NB)

Overall accuracy **83.1%**, macro-F1 0.826, AB recall **79.3%** — the lowest of all classifiers.

Worked example: PRF = 1,950 Hz (AB vs FF)



- FF: $z = (1950-1040)/185 = 4.92 \Rightarrow P \approx 4.7\times10^{-7}$ — very unlikely
- AB: $z = (1950-2280)/310 = -1.065 \Rightarrow P \approx 0.0023$ — far more plausible
- $\Rightarrow$ classifier strongly favours **AB**: correct!

Fails when: strong feature correlations (PRF and speed correlate for AB), non-Gaussian class-conditionals, important interactions.
Works for: text, fast baselines, small data, streaming.

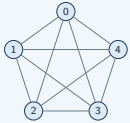
Multi-class Classification Strategies — Extending Binary Classifiers to 5 Classes

ONE vs REST (OvR)

f_{FF} vs ALL
f_{AB} vs ALL
f_{CM} vs ALL
f_{UV} vs ALL
f_{CI} vs ALL

5 classifiers; predict $\arg\max_c f_c(x)$

ONE vs ONE (OvO)



$C(C-1)/2 = 10$ classifiers; most wins

DIRECT MULTI-CLASS

Softmax / tree / RF / NB: native C-way output

best practice: use native when available

Trade-offs and IADS results

OvR: simple, parallel; but each sub-problem is imbalanced (1 vs 4). OvO: balanced sub-problems; but 10 classifiers for 5 classes (45 for 10). LR: OvR **89.7%**, OvO 90.1%, softmax 89.9%. SVM with OvO: **96.7%**. RF/tree: native.

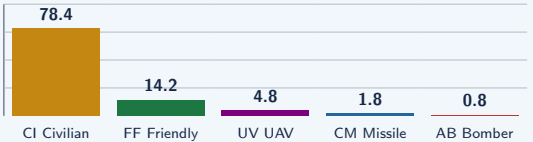
Calibrated probabilities

Raw scores $\neq$ probabilities. **Platt scaling** (SVM): sigmoid on scores; **isotonic regression**: monotone fit. A commander needs "80% confident this is AB", not just a label.

Reject option ($\theta = 0.6$)

If $\max P(y{=}c|x) < \theta \rightarrow$ refer to a human analyst. **4.2%** of signals sent to humans (98.7% human accuracy there); system + human combined: **99.1%**.

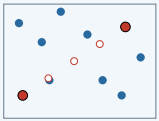
REAL IADS signal distribution vs our balanced set



Training on the real distribution: the model predicts "Civilian" for 92% of signals → 78.4%+14.2% = **92.6% accuracy**, yet **AB recall = 12.3%** (87.7% of bombers missed): clinically useless. Loss minimisation simply learns the imbalance.

Data pitfall: accuracy on imbalanced data is MEANINGLESS. Report precision, recall, F1 per class — a "98% accurate" model may have 0% recall on the most important class.

Four fixes



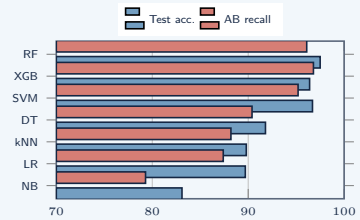
SMOTE: $x_{\text{new}} = x_i + \lambda(x_j - x_i)$

- **Oversample (SMOTE):** interpolate between real AB samples; AB 43 → 1,700 synthetic added; AB recall $12\% \rightarrow 89\%$
- **Undersample:** drop CI samples until balanced — fast but loses majority-class boundary information
- **Class-weighted loss:** $w_{\text{AB}} = N/(C \cdot N_{\text{AB}}) = 5780/(5 \cdot 43) = \mathbf{26.9}\times$ (vs 1.0) — most natural for RF/LR
- **Threshold shift:** classify AB if $P(\text{AB}|x) > 0.15$ instead of 0.5: more false alarms, higher recall — when FN cost $\gg$ FP cost

Algorithm Comparison — All Classifiers on the IADS Dataset

Algorithm	Train	Test	F1	AB rec.	Train (s)	Infer (ms)
Logistic Reg.	95.4%	89.7%	0.897	87.4%	0.8	0.09
k-NN (k=21)	91.8%	89.8%	0.894	88.2%	—	12.2
Decision Tree (d=10)	95.7%	91.8%	0.915	90.4%	0.5	0.15
Random Forest (B=500)	99.2%	**97.5%**	**0.975**	96.1%	12.4	2.8
XGBoost (M=400)	98.7%	96.4%	0.963	**96.8%**	18.6	**0.7**
SVM (RBF, C=100)	97.8%	96.7%	0.967	95.2%	38.4	4.1
Naive Bayes	89.1%	83.1%	0.826	79.3%	0.03	0.02

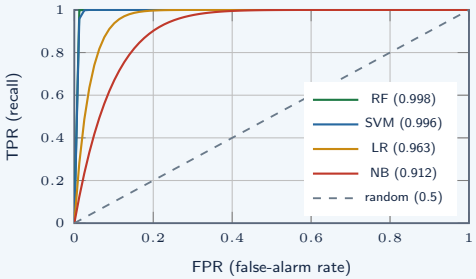
Balanced test set, $N = 1{,}275$ (15% of 8,500). SVM: 1,247 support vectors.



Winner: Random Forest (97.5%, AB recall 96.1%) for accuracy; XGBoost for fastest inference (0.7 ms); decision tree for interpretability.

NEVER use Naive Bayes for this task: AB recall 79.3% means 20.7% of adversary bombers are missed!

Beyond Accuracy — ROC & AUC for Defence AI



Schematic curves drawn to reproduce the reported AUCs (AB class, one-vs-rest); exact shapes are illustrative.

ROC curve & AUC

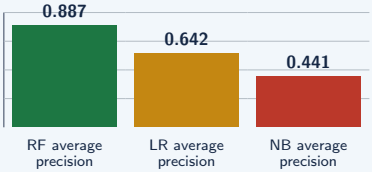
At each threshold θ: TPR $= TP/(TP+FN)$ (y), FPR $= FP/(FP+TN)$ (x). Ideal: top-left corner $(0,1)$; random: the diagonal, AUC $= 0.5$; AUC below the diagonal means an inverted classifier.

	RF	SVM	LR	DT	NB
AUC (AB, OvR)	**0.998**	0.996	0.963	0.940	0.912
TPR @ FPR 5%	**97.8%**	96.4%	88.3%	—	—

ROC-AUC is best when classes are balanced (lab conditions). On the real imbalanced stream, FPR hides how many alarms are false relative to true threats — precision–recall is more informative (next slide).

Precision–Recall, Average Precision and the F_β Score

AB class on real imbalanced data (0.8% AB)

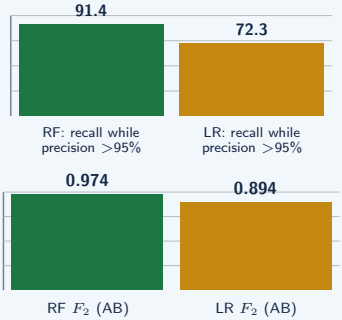


AVERAGE PRECISION AND F_β

$$AP = \sum_k (R_k - R_{k-1}) P_k, \qquad F_\beta = (1+\beta^2)\frac{PR}{\beta^2 P + R}$$

- $\beta > 1$ weights recall more (miss cost $>$ false-positive cost); $\beta = 2$ recommended for the IADS AB class

Operational metrics



Operational choice: use ROC-AUC for balanced (lab) classes; use the P–R curve / AP on real imbalanced data. For the IADS, F_2 **on the AB class** is the primary metric: recall matters twice as much as precision.

Satellite & SAR Image Interpretation — Four Tasks, One CNN Backbone

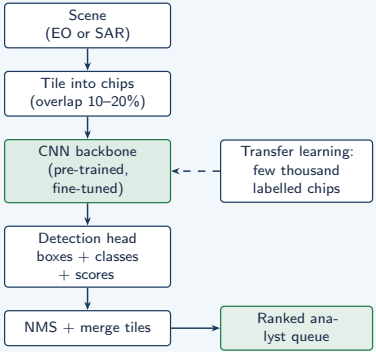
What the analyst needs

Task	Output	Example
Classification	one label per chip	"T-72-class tank"
Detection	box + label per object	all vehicles in 10 km^2
Segmentation	label per pixel	roads, buildings, water
Change detection	what differs between dates	new revetments since last week

Scale problem: a vehicle may be 10×20 pixels in a 30,000×30,000 scene. Scenes are cut into overlapping tiles; detections are merged across tile edges.

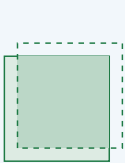
- **Two-stage** detectors (R-CNN family): propose regions, then classify — accurate, slower
- **One-stage** (YOLO, SSD, RetinaNet): boxes from a grid in one pass — real-time video
- **NMS** removes duplicate overlapping boxes

Detection pipeline

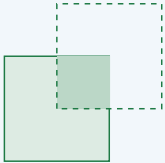


INTERSECTION OVER UNION

$$\text{IoU} = \frac{\text{area}(B_{\text{pred}} \cap B_{\text{true}})}{\text{area}(B_{\text{pred}} \cup B_{\text{true}})}$$



$\text{IoU} = 0.62$ **hit**

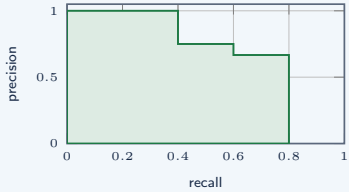


$\text{IoU} = 0.14$ **miss**

True box (solid) 40×40 px. Left: overlap $35^2 = 1{,}225$, union $1{,}975$. Right: overlap $20^2 = 400$, union $2{,}800$. A hit needs IoU ≥ 0.5.

Average precision: 5 vehicles, 8 detections

rank	1	2	3	4	5	6	7	8
hit?	**H**	**H**	**F**	**H**	**F**	**H**	**F**	**F**
precision	1.00	1.00	0.67	0.75	0.60	0.67	0.57	0.50
recall	0.2	0.4	0.4	0.6	0.6	0.8	0.8	0.8



AP $= 0.2(1) + 0.2(1) + 0.2(0.75) + 0.2(0.67) = \mathbf{0.68}$. One vehicle never found: recall stops at 0.8. **mAP** averages AP over classes (and IoU thresholds 0.5–0.95).

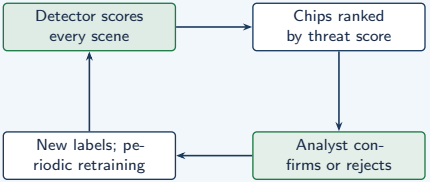
SAR Is Not a Photograph — Pitfalls and the Imagery-Triage Concept

Why overhead imagery is hard

- **SAR physics:** speckle noise, layover and radar shadow; a vehicle's signature changes with **look (aspect and depression) angle**
- **Distribution shift:** public SAR benchmarks (e.g. MSTAR, 10 vehicle classes) give near-perfect accuracy under matched conditions — and sharp drops when depression angle or vehicle configuration changes
- **Arbitrary orientation:** rotated boxes + rotation augmentation
- **Change detection:** naive differencing flags shadows, seasons and sensor changes; Siamese CNNs compare deep features of two dates
- **Deception:** decoys, camouflage nets, adversarial patterns target exactly these detectors

Defence: multi-sensor confirmation (EO + SAR + SIGINT); test on the deployment sensor and season, never only on the benchmark.

Operating concept: AI triage, analyst decides



Measure success operationally: analyst hours saved and threats found earlier *at a stated recall* — not benchmark mAP alone.

The same ideas recur in every science — and each has a defence application

AERONAUTICS / RADAR → PSYCHOLOGY
The ROC curve was born with WWII radar operators deciding "target or noise". Signal-detection theory (Peterson, Birdsall & Fox, 1954) then entered psychophysics (Tanner & Swets, 1954).
Defence: every IADS alert threshold is a point on an ROC curve.

BIOLOGY / MEDICINE
Cancer screening trades sensitivity against false recalls; the operating point is set by the cost of a missed case, not by accuracy.
Defence: recall-first thresholds for the rare Adversary-Bomber class (Lecture 2 case study).

ENGINEERING
Vibration signatures classify bearing and gearbox faults before failure — the basis of helicopter health & usage monitoring (HUMS).
Defence: fleet availability; "healthy vs incipient fault" with heavy class imbalance.

PSYCHOLOGY
Meehl (1954), *Clinical versus Statistical Prediction*: simple actuarial rules matched or beat expert judgement in most comparisons.
Defence: structured classifiers can outperform unaided gut calls on track ID — with a human still accountable.

STATISTICS
Fisher's iris data (1936) introduced linear discriminant analysis — ancestor of logistic regression and SVM boundaries.
Defence: separating radar-emitter classes using PRF and pulse width.

WARGAMING
Identifying OPFOR unit types from partial, fog-of-war observations; rare but critical classes (HQs, air-defence nodes) need per-class recall and calibrated confidence.
Use: kNN/RF on observed signatures.

Common thread: a classifier is a decision rule under uncertainty. Choose the threshold from the cost of each error type in *your* mission — radar operators, physicians and maintenance engineers all learned this the hard way.

Lecture 2 Summary — Six Classifiers, One Mission

LOGISTIC REGRESSION
Fast, interpretable (weights); linear boundaries only; baseline for all tasks.
Use: interpretable baseline or well-separated classes

kNN ($k = 21$)
No training; local structure; expensive inference $O(N)$; bias–variance tied to k.
Use: rare events; updatable in real time

DECISION TREE ($d = 10$)
Fully interpretable IF–THEN; axis-aligned boundaries; pruning prevents overfit.
Use: ROE compliance; commander explainability

RANDOM FOREST (winner)
97.5% accuracy, F1 0.975; robust, low overfit; feature importance for free.
Use: BEST overall for tabular defence classification

XGBOOST
96.4%; fastest inference 0.7 ms; sequential error correction; many hyper-parameters.
Use: real-time embedded systems

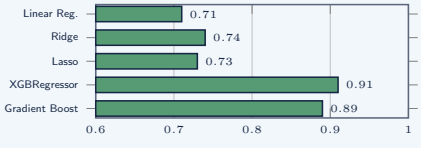
SVM (RBF)
96.7%; principled margin; sparse SVs (21.6%); training $O(N^2\text{–}N^3)$.
Use: small–medium datasets needing a margin guarantee

Lecture 3 topics

- Linear regression: OLS derivation (closed form)
- Regularised regression: Ridge (L2, shrinks), Lasso (L1, sparsity), Elastic Net
- Support vector regression: ε-insensitive loss; kernel SVR
- Gradient boosting for regression (XGBRegressor on logistics data)
- **Practical:** casualty estimation; force-effectiveness regression
- **Sensor fusion:** Kalman filter; ML-enhanced fusion; radar + IR + ELINT (3 sensors, 5 targets); late vs early fusion
- Prediction uncertainty & Bayesian regression

Assignment 2 is released with this lecture pack.

Preview: time-to-mission-completion ($N = 6{,}200$, 18 features)



Model	R^2	RMSE (h)
Linear	0.71	4.2
Ridge (λ=0.8)	0.74	3.9
Lasso (λ=0.3)	0.73	4.0
Gradient boost	0.89	2.3
XGBRegressor	**0.91**	**2.1**

Fusion preview

3-sensor (radar + IR + ELINT) track correlation: a single sensor vs a Kalman filter vs ML fusion — Lecture 3.